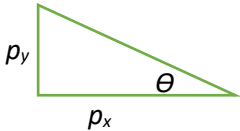
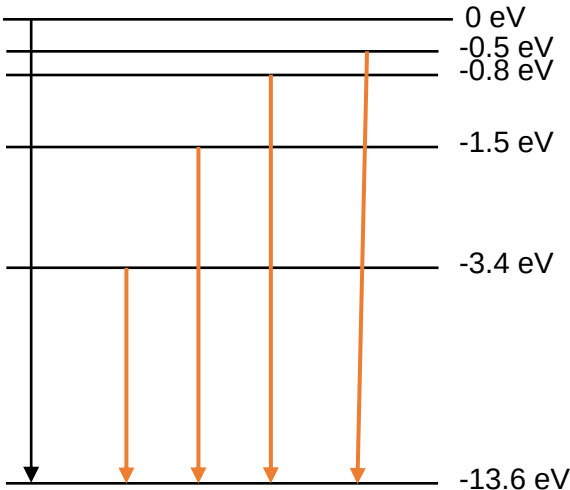


	<u>Due to different wavelengths of light having different fringe separations.</u>	M1
4ai	Maximum KE = $hf - 2.2\text{eV}$ = $9.4 \times 10^{-20} \text{ J}$	M1 A1
4aii	Stopping potential = $9.4 \times 10^{-20} / e = 0.587\text{V}$	A1
4aiii	<u>The values remain the same.</u> Intensity = Rate of Photons incident per unit area x Energy of 1 photon. Since the wavelength of the incident radiation is kept constant, <u>energy of the incident photon remains the same</u> , hence KE and Stopping potential are not affected. <u>Doubling the intensity will only double the rate of incident photons per unit area.</u>	A1 M1
4bi	It is <u>not possible to measure position or momentum</u> (or energy and time interval during which an object is in that state) <u>of an object precisely at the same time.</u> Note: Definition for Heisenberg position-momentum is if measurement of position is made with precision Δx and a simultaneous measurement of momentum in the x-direction is made with precision Δp , then the product of the two uncertainties can never be smaller than h (<i>Planck's constant</i>).	A1
4bii	Using De Broglie's relationship: $\lambda = h/p$ $= 6.63 \times 10^{-34} / 9.11 \times 10^{-31} \times 3.75 \times 10^6$ $= 0.1942 \text{ nm}$ $\sin \theta = \frac{\lambda}{D} = 3.88 \times 10^{-5}$ At small angles, $\sin \theta \approx \tan \theta$ Now $\tan \theta = \frac{p_y}{p_x} = 3.88 \times 10^{-5}$ $p_y = (9.11 \times 10^{-31})(3.75 \times 10^6)(3.88 \times 10^{-5})$ $= 1.33 \times 10^{-28} \text{ kg m s}^{-1}$ Therefore, maximum y-component of the momentum of an electron = $1.33 \times 10^{-28} \text{ kg m s}^{-1}$ 	M1 M1 M1 A0

4biii	$\Delta y \Delta p_y \geq h$ $\Delta y \geq \frac{h}{\Delta p_y} = \frac{6.63 \times 10^{-34}}{1.33 \times 10^{-28}}$ $\Delta y \geq 4.98 \times 10^{-6} \text{ m}$	M1 A1
4c	Applying de Broglie's wavelength, $\lambda = \frac{h}{p} = \frac{(6.63 \times 10^{-34})}{(9.11 \times 10^{-31})(1.26 \times 10^4)} = 5.78 \times 10^{-8} \text{ m}$ Applying $n\lambda = d \sin \theta$, For 2nd order maximum, $(1.6 \times 10^{-6}) \sin \theta_2 = 2 (5.78 \times 10^{-8})$, $\theta = 0.0723 \text{ rad}$	M1 A1
4di	$E_{95.0} = \frac{hc}{(95.0 \times 10^{-9})(1.6 \times 10^{-19})} = 13.1 \text{ eV}$ $E_{97.3} = \frac{hc}{(97.3 \times 10^{-9})(1.6 \times 10^{-19})} = 12.8 \text{ eV}$ $E_{102.6} = \frac{hc}{(102.6 \times 10^{-9})(1.6 \times 10^{-19})} = 12.1 \text{ eV}$ $E_{121.6} = \frac{hc}{(121.6 \times 10^{-9})(1.6 \times 10^{-19})} = 10.2 \text{ eV}$ Values of energy level calculated correctly (2 marks for all correct, 1 mark if 2 wrong, 0 marks for more than 2 wrong)	B2
4dii	 Arrows for emission lines to ground state drawn correctly. (2 marks for all correct, 1 mark if 2 wrong, 0 marks for more than 2 wrong). Minus 1 mark if no labelled clearly.	B2
5ai	Gold nucleus and alpha particle are positive charged. Thus the gold nucleus repels the	A1